SYSC 5108 W26 Assignment 2

1. Assume that the prior for q, the proportion of spam emails sent is a RV with uniform distribution. Let us denote by x the number of spam emails you receive in a random sample of n. To guess what the q is for you to receive these spam emails, you need to find the posterior distribution for $p(q|x)$. Which distribution will $p(q|x)$ be?

2. Given n independent sample points $x_i, i = 1, \cdots, n$ from a Laplace distribution with PDF $f(x) = \frac{\lambda}{2} e^{-\lambda|x|}$. Provide MLE estimators for the parameter λ.

3. A multivariate logistic regression model has been built to predict the propensity of shoppers to perform a repeat purchase of a free gift that they are given. The input features used by the model are the age of the customer, the socioeconomic band to which the customer belongs (a, b, or c), the average amount of money the customer spends on each visit to the shop, and the average number of visits the customer makes to the shop per week. This model is being used by the marketing department to determine who should be given the free gift. The weights in the trained model are shown in the following table. It is assumed that one hot coding is used for a, b, and c. Furthermore, if both b and c are 0, a will be used. Therefore, a is encoded as b = c =0.

Feature	Weight
Intercept (**w** [0])	-3.82398
AGE	-0.02990
SOCIOECONOMIC BAND B	-0.09089
SOCIOECONOMIC BAND C	-0.19558
SHOP VALUE	0.02999
SHOP FREQUENCY	0.74572

Use this model to make predictions for each of the following query instances.

ID	AGE	SOCIOECONOMIC BAND	SHOP FREQUENCY	SHOP VALUE
1	56	b	1.60	109.32
2	21	c	4.92	11.28
3	48	b	1.21	161.19
4	37	c	0.72	170.65
5	32	a	1.08	165.39

4. In building multivariate logistic regression models, it is recommended that all continuous input features be normalized to the range [-1,1]. The following table shows a data quality report for the dataset used to train the model described in the last question.

Feature	Count	% Miss.	Card.	Min.	1st Qrt.	Mean	Median	3rd Qrt.	Max.	Std. Dev.
AGE	5,200	6	40	18	22	32.7	32	32	63	12.2
SHOP FREQUENCY	5,200	0	316	0.2	1.0	2.2	1.3	4.3	5.4	1.6
SHOP VALUE	5,200	0	3,730	5	11.8	101.9	100.14	174.6	230.7	72.1

Feature	Count	% Miss.	Card.	Mode	Mode Count	Mode %	2nd Mode	2nd Mode Count	2nd Mode %
SOCIOECONOMIC BAND	5,200	8	3	a	2,664	51.2	b	1,315	25.3
REPEAT PURCHASE	5,200	0	2	no	2,791	53.7	yes	2,409	46.3

On the basis of the information in this report, all continuous features were normalized using range normalization, and any missing values were replaced using mean for imputating continuous features and mode for imputating categorical features. After applying these data preparation operations, a multivariate logistic regression model was trained to give the weights shown in the following table.

Feature	Weight
Intercept ($\mathbf{w}[0]$)	0.6679
AGE	-0.5795
SOCIOECONOMIC BAND B	-0.1981
SOCIOECONOMIC BAND C	-0.2318
SHOP VALUE	3.4091
SHOP FREQUENCY	2.0499

Use this model to make predictions for each of the query instances shown in the following table (question marks refer to missing values).

ID	AGE	SOCIOECONOMIC BAND	SHOP FREQUENCY	SHOP VALUE
1	38	a	1.90	165.39
2	56	b	1.60	109.32
3	18	c	6.00	10.09
4	?	b	1.33	204.62
5	62	?	0.85	110.50

5. Consider the training for logistic regression model. Assume $K = 2$, $y = 0, or\ 1$. The loss function is defined below.

$$J(w) = -y\log\big(\sigma(z)\big) - (1-y)\log\left(1-\sigma(z)\right)$$

where $z = \boldsymbol{w}^T\boldsymbol{x}$, $x_0 = 1$.
(a) Show that

$$\nabla_w J(w) = -\boldsymbol{x}\big(y - \sigma(z)\big)$$

(b) Now change the encoding for y to $y = -1, or\ 1$. Show the result again. Explain the impact of this change.

6. Consider the training for logistic regression model using a loss close to LSE. Assume $K = 2$, and we define a differentiable loss function through the errors in probabilities as below.

$$J(\beta) = \sum_{i=1}^{M}\left[I_1(y^{(i)})(1-p_1(z^{(i)}))^2 + I_2(y^{(i)})\big(p_1(z^{(i)})\big)^2\right]$$

where $z = \beta^T\boldsymbol{x}$, $x_0 = 1$. Find the gradient of the loss function and compare with the MLE discussed in class.

7. The following images are handwritten instances of the digits 0 and 1. The images are small, 8 pixels by 8 pixels, and each pixel contains a gray level from the range [0, 7].

Rather than use individual pixel values, which can lead to very high-dimensional feature vectors, a simpler way to represent images for use with regression models is to calculate a histogram for each image and use this as the feature vector instead. In this case the histograms simply count the frequency of occurrence of each possible gray level in each image. The table that follows shows the histograms for a small dataset of 10 images split between examples of digits 0 and 1.

ID	GL-0	GL-1	GL-2	GL-3	GL-4	GL-5	GL-6	GL-7	DIGIT
0	31	3	6	2	7	5	6	4	0
1	37	3	1	4	1	3	2	13	1
2	31	3	4	1	8	7	3	7	0
3	38	2	3	0	1	1	5	14	1
4	31	5	3	2	5	2	5	11	0
5	32	6	3	2	1	1	5	14	1
6	31	3	5	2	3	6	2	12	0
7	31	4	3	4	1	5	5	11	0
8	38	4	2	2	2	4	4	8	1
9	38	3	2	3	4	4	1	9	1

A logistic regression model has been trained to classify digits as either 0 or 1. The weights in this model are as follows:

Intercept	GL-0	GL-1	GL-2	GL-3	GL-4	GL-5	GL-6	GL-7
$\mathbf{w}[0]$	$\mathbf{w}[1]$	$\mathbf{w}[2]$	$\mathbf{w}[3]$	$\mathbf{w}[4]$	$\mathbf{w}[5]$	$\mathbf{w}[6]$	$\mathbf{w}[7]$	$\mathbf{w}[8]$
0.309	0.100	−0.152	−0.163	0.191	−0.631	−0.716	−0.478	−0.171

This model has been used to make predictions for the instances in the training set above. These predictions, and the related calculations required for calculating error and w[j] error values are shown in the following table.

					W[j] error								
ID	$\sigma(z_i)$	y_i	error	s.e.	w[0]	w[1]	w[2]	w[3]	w[4]	w[5]	w[6]	w[7]	w[8]
0	0.0001	0	-0.0001	?	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	?	1	0.8586	0.7373	0.1042	3.8562	0.3127	0.1042	0.4169	0.1042	0.3127	0.2084	1.3549
2	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	?
3	0.0565	1	0.9435	0.8902	?	1.9118	0.1006	0.1509	0.0000	0.0503	0.0503	0.2516	0.7044
4	?	0	-0.0018	0.0000	0.0000	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5	0.0256	1	?	0.9495	0.0243	0.7765	0.1456	0.0728	0.0485	0.0243	0.0243	0.1213	0.3397
6	?	0	-0.0013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
7	0.0045	0	?	0.0000	0.0000	-0.0006	?	-0.0001	-0.0001	0.0000	-0.0001	-0.0001	-0.0002
8	0.0209	1	0.9791	?	0.0200	0.7598	0.0800	0.0400	0.0400	0.0400	0.0800	0.0800	0.1600
9	0.0292	1	0.9708	0.9425	0.0275	1.0447	0.0825	0.0550	0.0825	0.1100	0.1100	0.0275	0.2474

In the table, the w[j] errors are calculated as $-\frac{\partial \mathrm{L(W)}}{\partial \mathrm{W(j)}} = \big(y_i - \sigma(z_i)\big)\sigma(z_i)\big(1 - \sigma(z_i)\big)x_i[j]$ which is the gradient for $W[j]$ estimated using least squared error (LSE) instead of MLE. $x_i[j]$ is the ith feature of jth sample point.

(a) Some of the model predictions are missing in the preceding table (marked with a ?). Calculate these.

(b) Some of the Error and Squared Error values are missing in the preceding table (marked with a ?). Calculate these values.

(c) Some of the w[j] error values are missing in the preceding table (marked with a?). Calculate these.

(d) Calculate a new set of weights for this model using a learning rate of 0.01 with LSE.

(e) The following table shows handwritten examples of the digits 7 and 8 and their corresponding histogram values.

	GL-0	GL-1	GL-2	GL-3	GL-4	GL-5	GL-6	GL-7
	35	1	5	4	5	2	4	8
	30	6	2	0	5	4	4	13

i. Calculate the output of the model (using the updated weights calculated in the previous part) for these two instances.

ii. Comment on the appropriateness of these outputs.

(f) Now calculate new weights using MLE and one sample point ID = 0 only instead of step (d) and repeat step (e). Use the same learning rate.